

Lecture 14

IMPACT

Dr. Hiru Purushothaman Hirudayanathan
Faculty / Mechanical Engineering
Room No 20, Visvesvaraya Complex
National Institute of Technology Goa
Phone No: 8754097380

IMPACT

Today's Objectives:

Students will be able to:

1. Understand and analyze the mechanics of impact.
2. Analyze the motion of bodies undergoing a collision, in both central and oblique cases of impact.



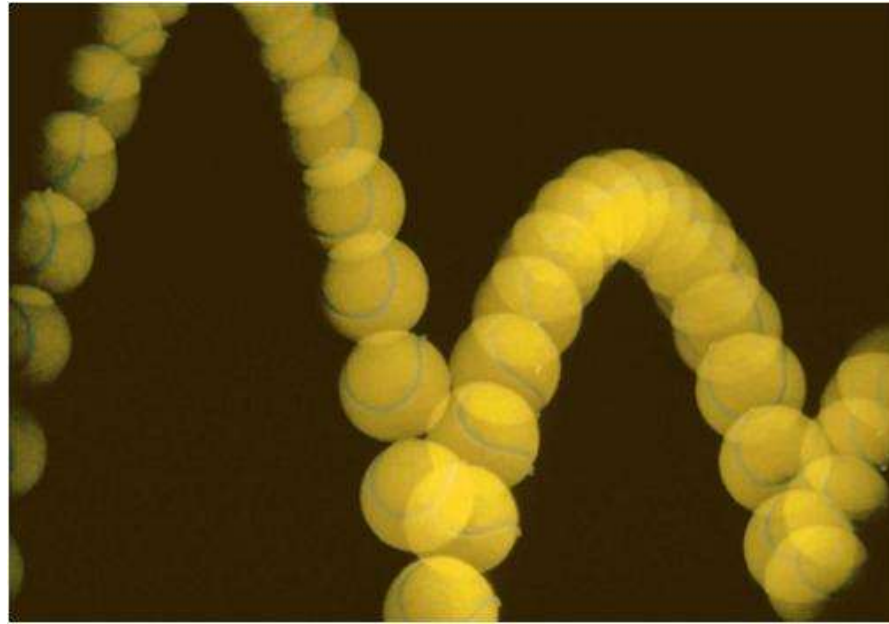
In-Class Activities:

- Reading Quiz
- Applications
- Central Impact
- Coefficient of Restitution
- Oblique Impact
- Example Problem
- Concept Quiz
- Group Problem Solving
- Attention Quiz

READING QUIZ

1. When the motion of one or both of the particles is at an angle to the line of impact, the impact is said to be _____
A) central impact. B) oblique impact.
C) major impact. D) None of the above.
2. The ratio of the restitution impulse to the deformation impulse is called _____
A) impulse ratio. B) restitution coefficient.
C) energy ratio. D) mechanical efficiency.

APPLICATIONS

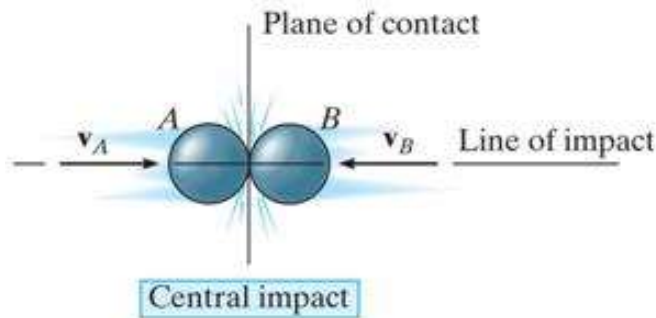


The quality of a tennis ball is measured by the height of its bounce. This can be quantified by the **coefficient of restitution** of the ball.

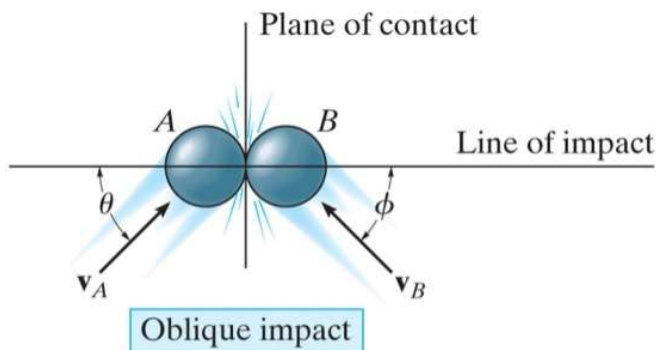
If the height from which the ball is dropped and the height of its resulting bounce are known, how can we determine the coefficient of restitution of the ball?

IMPACT (Section 15.4)

Impact occurs when two bodies collide during a very **short time period**, causing large impulsive forces to be exerted between the bodies. Common examples of impact are a hammer striking a nail or a bat striking a ball. The **line of impact** is a line through the **mass centers** of the colliding particles. In general, there are **two** types of impact:



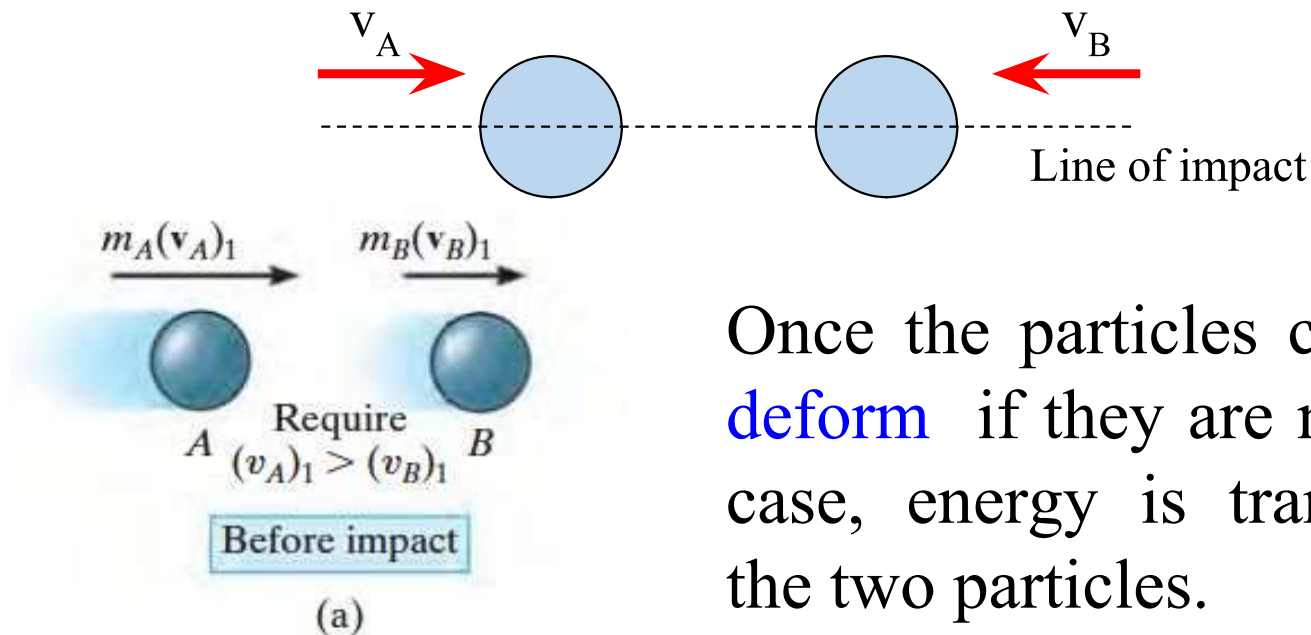
Central impact occurs when the directions of motion of the two colliding particles are along the line of impact.



Oblique impact occurs when the direction of motion of one or both of the particles is at an angle to the line of impact.

CENTRAL IMPACT

Central impact happens when the velocities of the two objects are along the line of impact (recall that the line of impact is a line through the particles' mass centers).

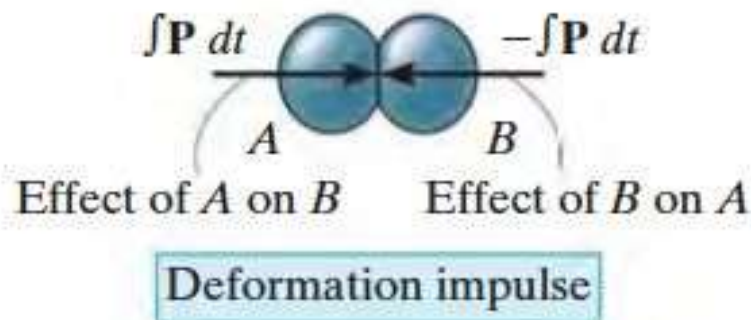


Once the particles contact, they may **deform** if they are non-rigid. In any case, energy is transferred between the two particles.

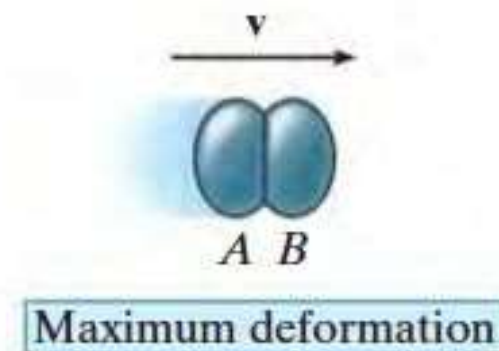
There are two primary equations used when solving impact problems.

CENTRAL IMPACT (continued)

- The particles have the initial momenta shown in Fig. 15–14*a*. Provided $(v_A)_1 > (v_B)_1$, collision will eventually occur.
- During the collision the particles must be thought of as *deformable* or nonrigid. The particles will undergo a *period of deformation* such that they exert an equal but opposite deformation impulse $\int \mathbf{P} dt$ on each other, Fig. 15–14*b*.
- Only at the instant of *maximum deformation* will both particles move with a common velocity $\mathbf{v}$, since their relative motion is zero, Fig. 15–14*c*.



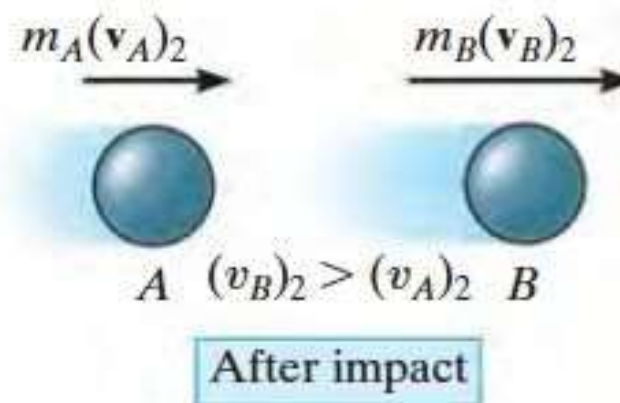
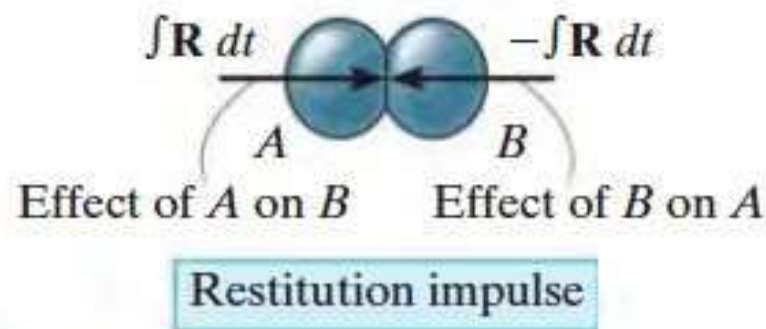
(b)



(c)

CENTRAL IMPACT (continued)

- Afterward a *period of restitution* occurs, in which case the particles will either return to their original shape or remain permanently deformed. The equal but opposite *restitution impulse* $\int \mathbf{R} dt$ pushes the particles apart from one another, Fig. 15–14d. In reality, the physical properties of any two bodies are such that the deformation impulse will *always be greater* than that of restitution, i.e., $\int P dt > \int R dt$.
- Just after separation the particles will have the final momenta shown in Fig. 15–14e, where $(v_B)_2 > (v_A)_2$.



CENTRAL IMPACT (continued)

In most problems, the initial velocities of the particles, $(v_A)_1$ and $(v_B)_1$, are known, and it is necessary to determine the final velocities, $(v_A)_2$ and $(v_B)_2$. So, the **first** equation used is the **conservation of linear momentum**, applied along the line of impact.

$$(m_A v_A)_1 + (m_B v_B)_1 = (m_A v_A)_2 + (m_B v_B)_2$$

This provides one equation, but there are usually two unknowns, $(v_A)_2$ and $(v_B)_2$. Another equation is needed. The **principle of impulse and momentum** is used to develop this equation, which involves the **coefficient of restitution, or e** .

CENTRAL IMPACT (continued)

The **coefficient of restitution, e** , is the ratio of the particles' **relative separation velocity** after impact, $(v_B)_2 - (v_A)_2$, to the particles' **relative approach velocity** before impact, $(v_A)_1 - (v_B)_1$. The coefficient of restitution is also an indicator of the energy lost during the impact.

The equation defining the coefficient of restitution, e , is

$$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$$

If a value for e is specified, this relation provides the second equation necessary to solve for $(v_A)_2$ and $(v_B)_2$.

COEFFICIENT OF RESTITUTION

In general, e has a value between zero and one.

The two limiting conditions can be considered:

- Elastic impact ($e = 1$): In a perfectly elastic collision, no energy is lost, and the relative separation velocity equals the relative approach velocity of the particles. In practical situations, this condition cannot be achieved.
- Plastic impact ($e = 0$): In a plastic impact, the relative separation velocity is zero. The particles stick together and move with a common velocity after the impact.

Some **typical values** of e are:

Steel on steel: 0.5 – 0.8 Wood on wood: 0.4 – 0.6

Lead on lead: 0.12 – 0.18 Glass on glass: 0.93 – 0.95

IMPACT: ENERGY LOSSES

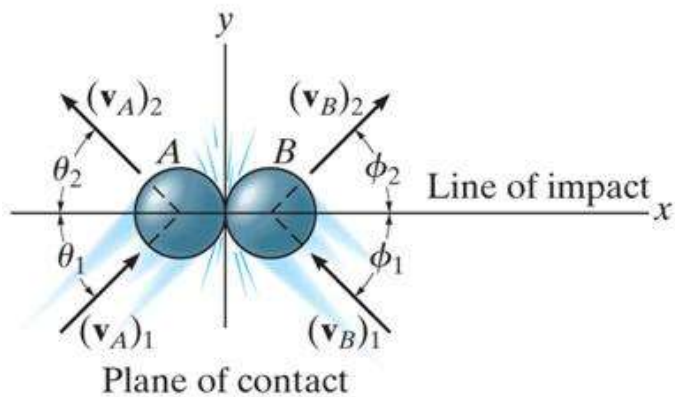
Once the particles' velocities before and after the collision have been determined, the **energy loss** during the collision can be calculated on the basis of the **difference in the particles' kinetic energy**. The energy loss is

$$\sum U_{1-2} = \sum T_2 - \sum T_1 \quad \text{where } T_i = 0.5m_i(v_i)^2$$

During a collision, some of the particles' initial kinetic energy will be lost in the form of heat, sound, or due to localized deformation.

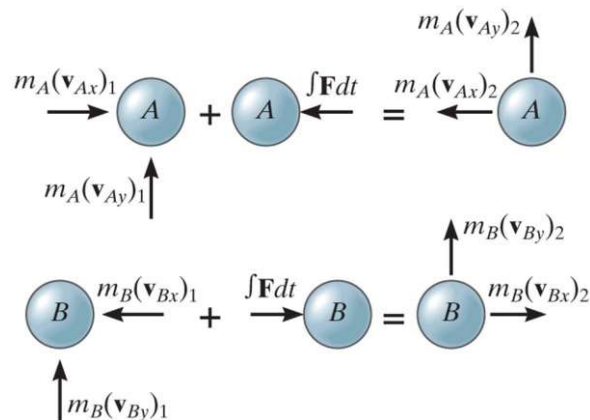
In a **plastic collision** ($e = 0$), the energy lost is a maximum, although the energy of the combined masses does not necessarily go to zero.

OBLIQUE IMPACT



In an **oblique impact**, one or both of the particles' motion is at an angle to the line of impact. Typically, there will be four unknowns: the **magnitudes** and **directions** of the final velocities.

The four equations required to solve for the unknowns are:



Conservation of momentum and the coefficient of restitution equation are applied **along** the line of impact (x-axis):

$$m_A(v_{Ax})_1 + m_B(v_{Bx})_1 = m_A(v_{Ax})_2 + m_B(v_{Bx})_2$$

$$e = [(v_{Bx})_2 - (v_{Ax})_2] / [(v_{Ax})_1 - (v_{Bx})_1]$$

Momentum of each particle is conserved in the direction **perpendicular** to the line of impact (y-axis):

$$m_A(v_{Ay})_1 = m_A(v_{Ay})_2 \quad \text{and} \quad m_B(v_{By})_1 = m_B(v_{By})_2$$

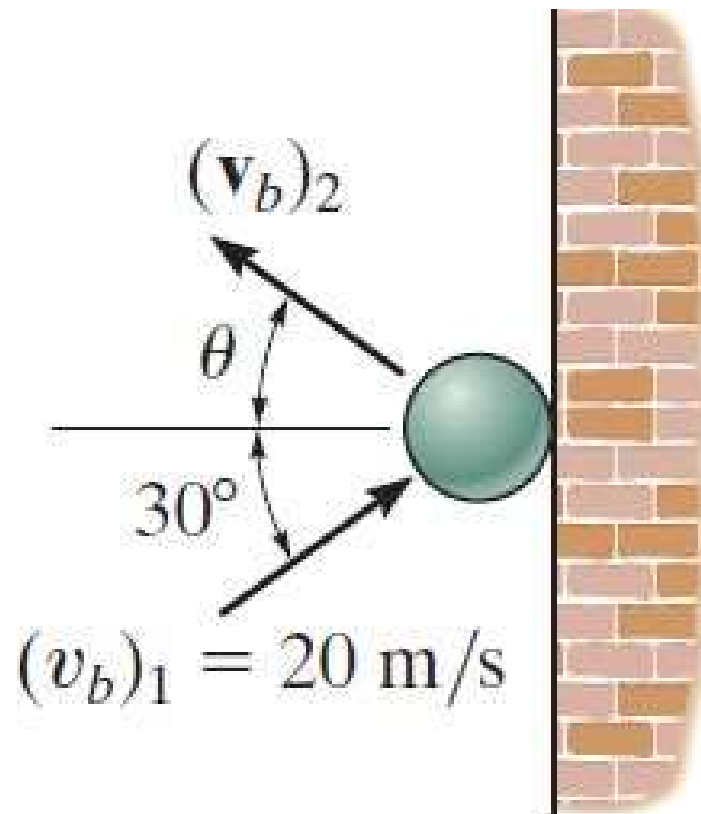
PROCEDURE FOR ANALYSIS

- In most impact problems, the initial velocities of the particles and the coefficient of restitution, e , are known, with the final velocities to be determined.
- Define the x-y axes. Typically, the **x-axis** is defined **along** the line of impact and the **y-axis** is in the plane of contact **perpendicular** to the x-axis.
- For both **central and oblique** impact problems, the following equations apply **along** the line of impact (x-dir.):
$$\sum m(v_x)_1 = \sum m(v_x)_2 \quad \text{and} \quad e = [(v_{Bx})_2 - (v_{Ax})_2] / [(v_{Ax})_1 - (v_{Bx})_1]$$
- For **oblique** impact problems, the following equations are also required, applied **perpendicular** to the line of impact (y-dir.):

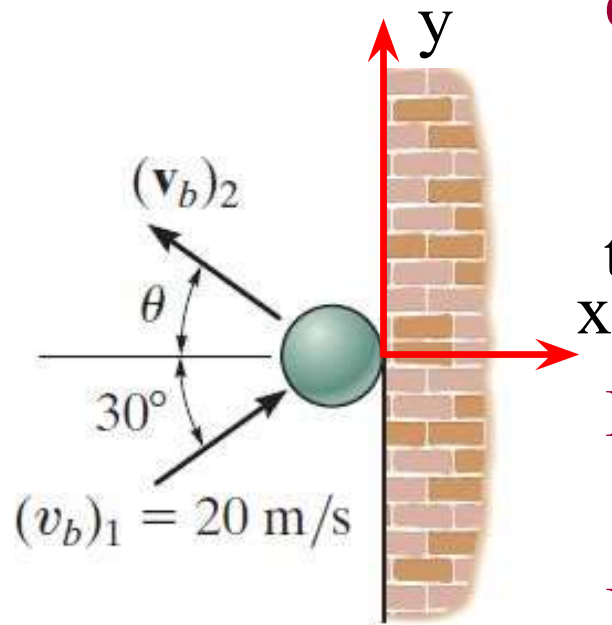
$$m_A(v_{Ay})_1 = m_A(v_{Ay})_2 \quad \text{and} \quad m_B(v_{By})_1 = m_B(v_{By})_2$$

EXAMPLE

The ball strikes the smooth wall with a velocity of $(v_b)_1 = 20$ m/s. If the coefficient of restitution between the ball and the wall is $e = 0.75$, determine the velocity of the ball just after the impact. **(Problem No. F15-16)**



EXAMPLE



Given: The ball strikes the smooth wall with a velocity $(v_b)_1 = 20 \text{ m/s}$. The coefficient of restitution between the ball and the wall is $e = 0.75$.

Find: The velocity of the ball just after the impact.

Plan:

The collision is an **oblique impact**, with the line of impact perpendicular to the plane (through the relative centers of mass).

Thus, the coefficient of restitution applies perpendicular to the wall and the momentum of the ball is conserved along the wall.

EXAMPLE (continued)

Solution:

Solve the impact problem by using x-y axes defined along and perpendicular to the line of impact, respectively:

The **momentum** of the ball is **conserved in the y-dir**:

$$\begin{aligned} m(v_b)_1 \sin 30^\circ &= m(v_b)_2 \sin \theta \\ (v_b)_2 \sin \theta &= 10 \text{ m/s} \end{aligned} \quad (1)$$

The **coefficient of restitution** applies in the **x-dir**:

$$\begin{aligned} e &= [0 - (v_{bx})_2] / [(v_{bx})_1 - 0] \\ \Rightarrow 0.75 &= [0 - (-v_b)_2 \cos \theta] / [20 \cos 30^\circ - 0] \\ \Rightarrow (v_b)_2 \cos \theta &= 12.99 \text{ m/s} \end{aligned} \quad (2)$$

Using Eqs. (1) and (2) and solving for the velocity and θ yields:

$$\begin{aligned} (v_b)_2 &= \sqrt{12.99^2 + 10^2} = 16.4 \text{ m/s} \\ \theta &= \tan^{-1}(10/12.99) = 37.6^\circ \end{aligned}$$

Using Eqs. (1) and (2) and solving for the velocity and θ yields:

$$(v_b)_2 = \sqrt{12.99^2 + 10^2} = 16.4 \text{ m/s}$$

$$\theta = \tan^{-1}(10/12.99) = 37.6^\circ$$

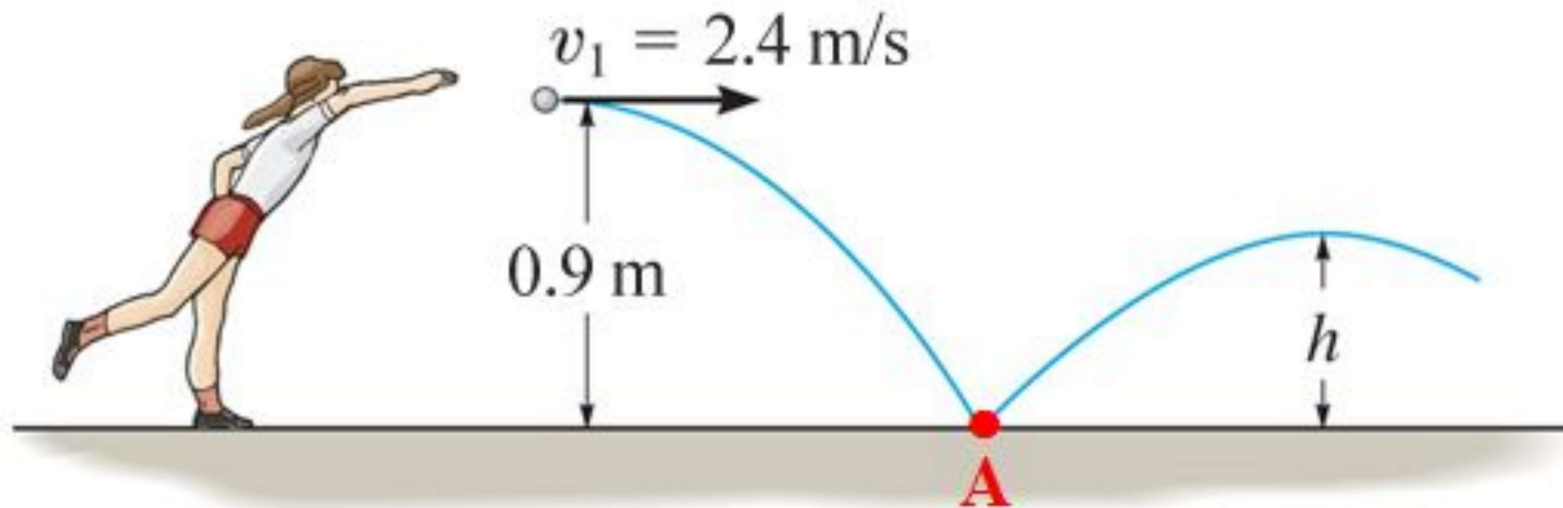


CONCEPT QUIZ

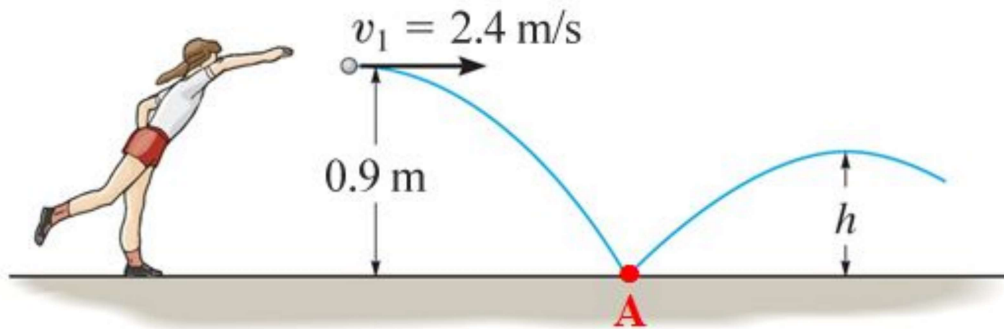
1. Two balls impact with a coefficient of restitution of 0.79. Can one of the balls leave the impact with a kinetic energy greater than before the impact?
A) Yes B) No
C) Impossible to tell D) Don't pick this one!
2. Under what condition is the energy lost during a collision maximum?
A) $e = 1.0$ B) $e = 0.0$
C) $e = -1.0$ D) Collision is non-elastic.

GROUP PROBLEM SOLVING

The girl throws the ball with a horizontal velocity $v_1 = 2.4 \text{ m/s}$. If the coefficient of restitution between the ball and the ground is 0.8, determine (a) the velocity of the ball just after it rebounds from the ground and (b) the maximum height to which the ball rises after the first bounce.



GROUP PROBLEM SOLVING



Given: The girl throws the ball with a velocity of $v_1 = 2.4 \text{ m/s}$. The coefficient of restitution between the ball and the hard ground is $e = 0.8$.

Find: The rebounding velocity of the ball at A, the maximum h .

- Plan:**
- 1) Determine the speed of the ball just before hitting the ground using projectile motion.
 - 2) Apply the coefficient of restitution in the y-dir motion, and the conservation of momentum in the x-dir motion..
 - 3) Use kinematics equations to find h .

GROUP PROBLEM SOLVING (continued)

Solution:

- 1) By considering the vertical motion of the falling ball, we have:

$$(v_{Ay})^2 = (v_{1y})^2 + 2 a_c (s_{Ay} - s_{1y})$$

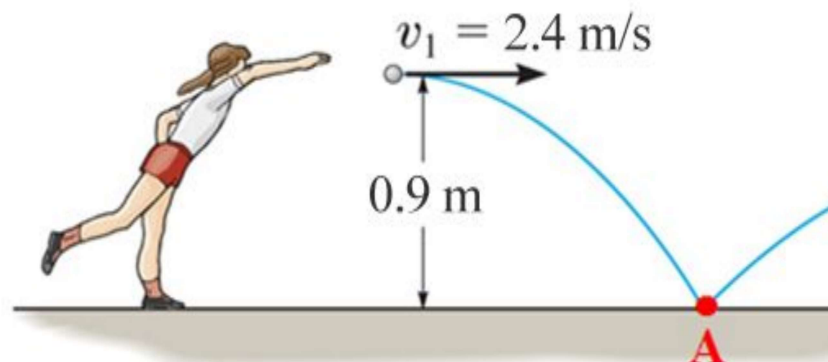
$$(v_{Ay})^2 = 0 + 2 (-9.81) (0 - 0.9)$$

$$v_{Ay} = -4.202 \text{ m/s} = 4.202 \text{ m/s} \downarrow$$

- 2) Apply the coefficient of restitution in the y -dir to determine the velocity of the ball just after it rebounds from the ground.

$$e = \frac{V_{g2} - (V_{Ay})_2}{(V_{Ay})_1 - V_{g1}} \Rightarrow 0.8 = \frac{0 - (V_{Ay})_2}{-4.202 - 0}$$

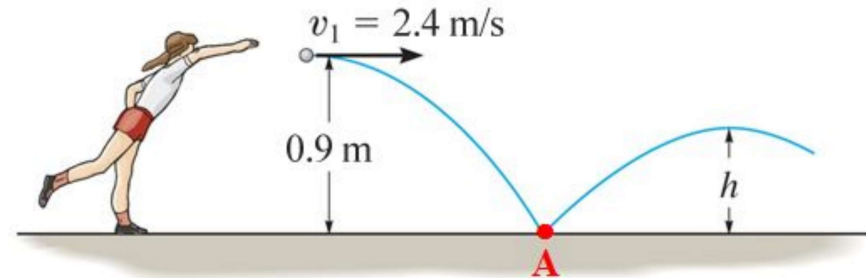
$$(v_{Ay})_2 = 3.362 \text{ m/s} \uparrow$$



GROUP PROBLEM SOLVING (continued)

Apply conservation of momentum to the system in the x-dir :

$$+ \rightarrow m (v_{1x}) = m (v_{Ax})$$
$$(v_{Ax}) = 2.4 \text{ m/s} \rightarrow$$



Therefore, the rebounding velocity of the ball at A is

$$(\mathbf{v}_A)_2 = (2.4 \mathbf{i} + 3.362 \mathbf{j}) \text{ m/s}$$

3) Kinematics equations: the vertical motion of the ball after it rebounds from the ground is described by:

$$(v_{2y})^2 = (v_{Ay})^2 + 2 a_c (s_{2y} - s_{Ay}) \text{ and } v_{2y} \text{ is zero at height } h$$

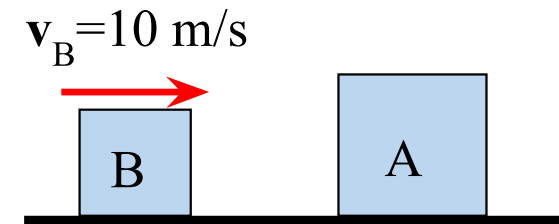
$$0^2 = (3.362)^2 + 2 (-9.81) (h - 0)$$

$$\underline{h = 0.576 \text{ m}}$$

ATTENTION QUIZ

1. Block B (1 kg) is moving on the smooth surface at 10 m/s when it squarely strikes block A (3 kg), which is at rest. If the velocity of block A after the collision is 4 m/s to the right, $(v_B)_2$ is

- A) 2 m/s $\rightarrow$ B) 7 m/s $\leftarrow$
C) 7 m/s $\rightarrow$ D) 2 m/s $\leftarrow$



End of the Lecture
Let The Learning Continue